UACJ

*La Universidad Autónoma de Ciudad Juárez*




ENCUENTRO DE

# JÓVENES INVESTIGADORES

en el Estado de Chihuahua, 2024

Modalidad Presencial



# *Spatial relationship between intentional homicides and the socioeconomic situation of the homicide perpetrator.*

Authors:
Mechelle Annet Hernández Elizarraras
Jaime García de la Rosa

Field of knowledge: Social Sciences / Economics

Septiembre / September, 202

# OVERVIEW

*Ciudad Juárez is described by Payán (2017) as the collapsed city, due to the presence of violence and crime in recent decades.*

*One of the crimes with the greatest impact on Juárez society is intentional homicide, which has direct effects on the well-being of its inhabitants.*

*Authors such as Bass Zavala (2013) and Barajas Bustillos and García de la Rosa (2016) have investigated the causes that give rise to crime and social precariousness. Among them are high migration and premature urban growth.*

*The homicides in the city have a specific spatial pattern, in which certain areas where they cluster can be identified.*

*Likewise, areas identified as generators of crime have been recognized, characterized by social deprivation and low crime rates.*

*The general purpose of the study is to contribute to the analysis of this phenomenon based on the social and economic conditions of the perpetrator, that is, of whoever is the material actor of the crime.*

# THEORETICAL FRAMEWORK

| Theory | Autores/as / Authors | Used variables | Results |
|---|---|---|---|
| Economic theories | Becker (1968)<br>Ehrlich (1973)<br>Romero Pérez (2012)<br>Enamorado et al. (2016) | - Racionalidad económica. / Economic rationality.<br>- Rendimientos esperados. / Expected returns.<br>- Beneficios legales. / Legal benefits.<br>- Probabilidad de ser atrapado/a. / Probability of being caught.<br>- Castigo. / Punishment. | An optimal level of crime.<br>Inequality and income conditions are positively related to both property crimes and crimes against physical integrity. |
| Ecological theories of crime | García et al. (2019)<br>Boehme (2020)<br>Loureiro et al. (2009) | - Motivaciones. / Motivations.<br>- Características ambientales. / Environmental characteristics.<br>- Características sociales de los lugares. / Social characteristics of places. | Areas defined as “criminogenic”.<br>Crimes not random in space.<br>Areas that attract crimes.<br>Areas that generate offenders. |
| Frustration theory and social disorganization theory | Merton (1938)<br>Shaw y McKay (1942)<br>Carrillo y García (2021) | - Desarrollo de emociones negativas, frustración. / Development of negative emotions, frustration.<br>- Sentimiento de fracaso e inferioridad. / Feeling of failure and inferiority.<br>- Falta de organización social. / Lack of social organization.<br>- Variables sociodemográficas y culturales. / Sociodemographic and cultural variables. | El comportamiento delictivo tendrá una mayor probabilidad de ocurrir si el individuo se enfrenta a una sociedad donde la desigualdad es más elevada. / Criminal behavior is more likely to occur if the individual faces a society where inequality is higher.<br>Zonas con alta criminalidad, debido a la ausencia de un control social. / Areas with high crime, due to the absence of social control.<br>Falta de eficiencia en el control social. / Lack of efficiency in social control. |

# OBJETIVO (S) / OBJECTIVE(S)

Objetivo general / General objective:

- Analizar las condiciones socioeconómicas de los lugares de residencia de los victimarios de homicidio en Ciudad Juárez, como el rezago educativo, bajos niveles de ingreso, la marginación y el rezago social.
- *To analyze the socioeconomic conditions of the places of residence of homicide perpetrators in Ciudad Juárez, such as educational lag, low income levels, marginalization and social lag.*

Objetivos específicos / Specific objectives:

- Construir las variables de carácter económico y social en corroboración con la literatura, que tengan relevancia en el contexto espacial y socioeconómico de Ciudad Juárez.
- *To build economic and social variables, supported by the literature, that are relevant to the spatial and socioeconomic context of Ciudad Juárez.*
- Determinar a través de estadística espacial y geoestadística la probabilidad de que una persona cometa un homicidio a partir del conjunto de características socioeconómicas de los lugares.
- *To determine, through spatial statistics and geostatistics, the probability that a person commits a homicide based on the set of socioeconomic characteristics of the places.*

# JUSTIFICATION

*The strategies and legislation implemented by the State seem to have had no significant effects in reducing the homicide rate.*

*In 2023, Ciudad Juárez closed with a total of 1,163 homicides (González, 2023), a figure that kept it among the ranking of the 10 most violent cities in Mexico.*

*In economic terms, negative externalities are generated that closely involve the well-being of the Juárez population, as well as an unceasing dynamism of illegal markets. Likewise, unequal circumstances are found in certain areas of the city, where the ties to crime and the offense of homicide grow stronger.*

# METHODOLOGY

*In order to verify that the socioeconomic conditions of the places in the city have a non-random or unequal distribution, an analysis was carried out using descriptive statistical methods, such as standard deviation and weighted means, as well as the use of spatial statistical inference through georeferencing. In this way, urban social mobility was also studied.*

## TECHNIQUE

For the interpolation calculation, the Kriging formula is determined as follows (Cervera Gómez, 2015, p. 34):

$$\hat{Z}(s_0) = \sum_{i=1}^{N} \lambda_i Z(s_i)$$

**Where:**

$Z(s_i)$= **the value measured at location** $_i$

$\lambda_i$= **an unknown weight for the value measured at location** $_i$

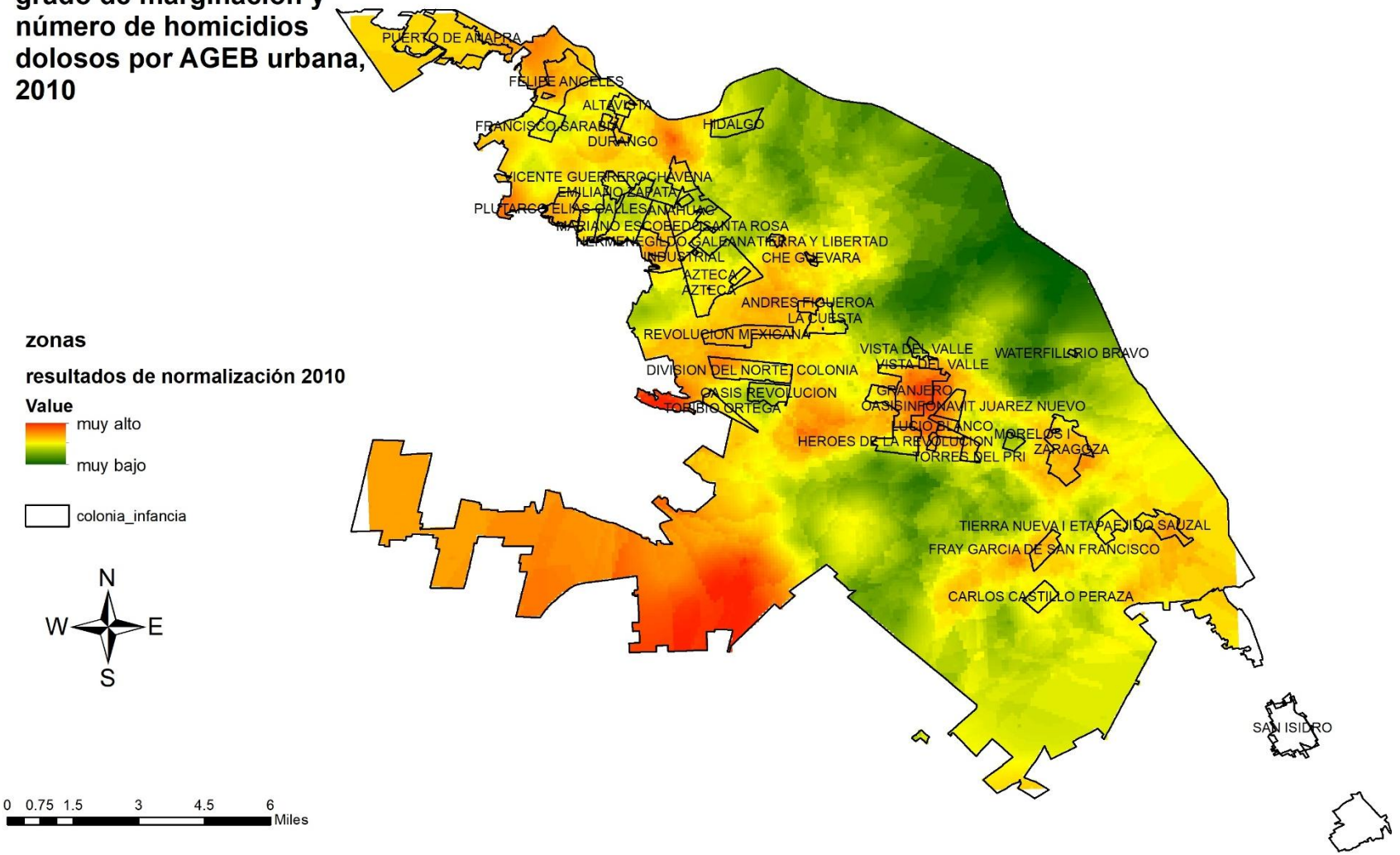
$s_0$= **the prediction location**

**N= the number of measured values**

*What this technique yields is a value that can be taken as the probability that a person commits the crime of homicide per urban AGEB (census area).*

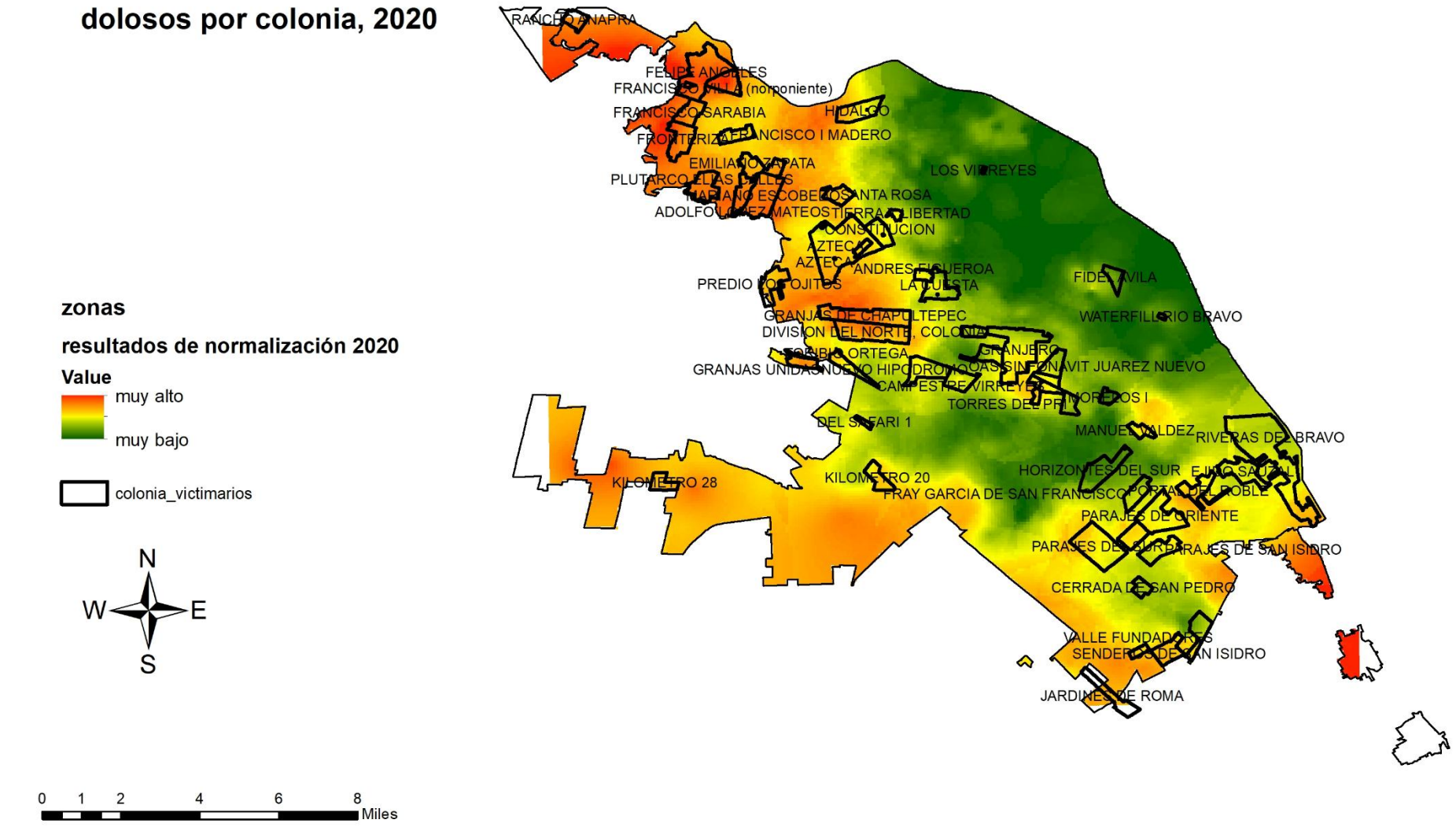
# RESULTS

**Resultados de normalización: grado de marginación y número de homicidios dolosos por AGEB urbana, 2010**

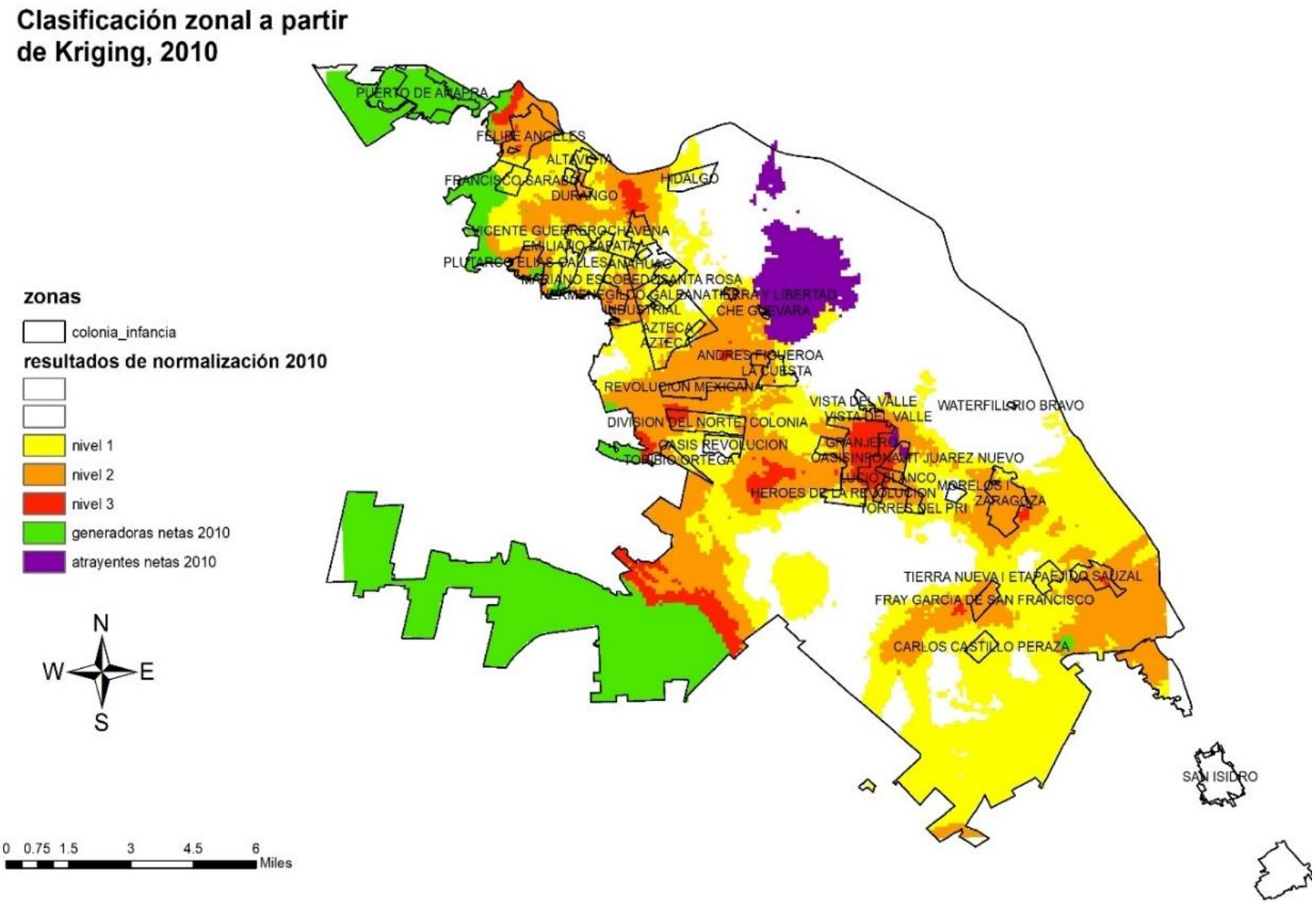


# RESULTS

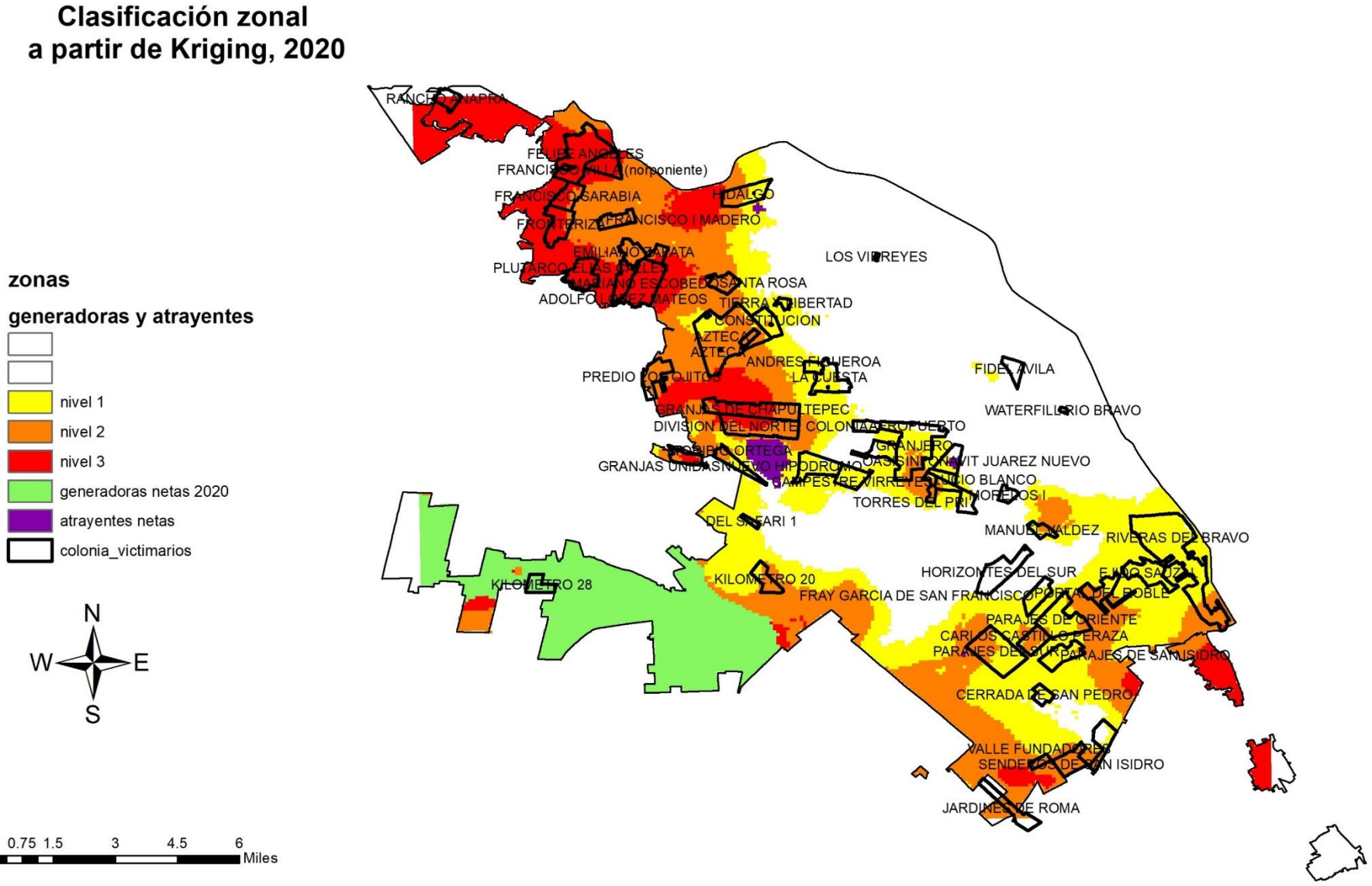
## Resultados de normalización: grado de marginación y número de homicidios dolosos por colonia, 2020



# RESULTS


Clasificación zonal a partir de Kriging, 2010
zonas
colonia_infancia
resultados de normalización 2010
nivel 1
nivel 2
nivel 3
generadoras netas 2010
atrayentes netas 2010
N
W
E
S
0 0.75 1.5 3 4.5 6
Miles
PUERTO DE ANAPRA
FELIPE ANGELES
ALTAVISTA
FRANCISCO SARABIA
DURANGO
HIDALGO
VICENTE GUERRERO
CHAVENA
EMILIANO ZAPATA
PLUTARCO ELIAS CALLES
ANAHUAC
MARIANO ESCOBEDO
SANTA ROSA
HERMENEGILDO GALEANA
TIERRA Y LIBERTAD
INDUSTRIAL
CHE GUEVARA
AZTECA
AZTECA
ANDRES FIGUEROA
LA CUESTA
REVOLUCION MEXICANA
VISTA DEL VALLE
VISTA DEL VALLE
WATERFILL
RIO BRAVO
DIVISION DEL NORTE
COLONIA
OASIS REVOLUCION
GRANJERO
TOMBIO ORTEGA
OASIS INN
JUAREZ NUEVO
LUCIO BLANCO
MORELOS I
HEROES DE LA REVOLUCION
ZARAGOZA
TORRES DEL PRI
TIERRA NUEVA I ETAPA
EJIDO SAUZAL
FRAY GARCIA DE SAN FRANCISCO
CARLOS CASTILLO PERAZA
SAN ISIDRO

# RESULTS


Clasificación zonal
a partir de Kriging, 2020
zonas
generadoras y atrayentes
nivel 1
nivel 2
nivel 3
generadoras netas 2020
atrayentes netas
colonia_victimarios
N
W
E
S
0 0.75 1.5 3 4.5 6
Miles
RANCHO ANAPRA
FELIPE ANGELES
FRANCISCO VILLA (norponiente)
FRANCISCO SARABIA
HIDALGO
FRONTERIZA
FRANCISCO I MADERO
EMILIANO ZAPATA
LOS VIRREYES
PLUTARCO ELIAS CALLES
MARIANO ESCOBEDO
SANTA ROSA
ADOLFO LOPEZ MATEOS
TIERRA Y LIBERTAD
CONSTITUCION
AZTECA
AZTECA
ANDRES FIGUEROA
PREDIO LOS ROJITOS
LA CUESTA
FIDEL AVILA
GRANJAS DE CHAPULTEPEC
WATERFILL
RIO BRAVO
DIVISION DEL NORTE
COLONIA AEROPUERTO
GRANJERO
ROBERTO ORTEGA
GRANJAS UNIDAS
NUEVO HIPODROMO
OASIS INFONAVIT JUAREZ NUEVO
CAMPESTRE VIRREYES
RANCHO BLANCO
MORELOS I
TORRES DEL PRI
DEL SAFARI 1
MANUEL VALDEZ
RIVERAS DEL BRAVO
HORIZONTES DEL SUR
KILOMETRO 28
KILOMETRO 20
FRAY GARCIA DE SAN FRANCISCO
PORTAL DEL ROBLE
PARAJES DE ORIENTE
CARLOS CASTILLO PERAZA
PARAJES DEL SUR
PARAJES DE SAN ISIDRO
CERRADA DE SAN PEDRO
VALLE FUNDADORES
SENDEROS DE SAN ISIDRO
JARDINES DE ROMA

## CONCLUSIONS AND CONTRIBUTIONS

*In most cases, both variables are present for the two analysis periods; that is, the areas of residence of perpetrators are mostly both generators and attractors.*

*Conditions of precariousness persist in the west and southeast of the city, differing from the rest by maintaining medium and high levels of marginalization. Likewise, homicidal violence is redirected toward these places.*

*The distribution of dwellings is located over the area that concentrates high marginalization indices for both periods.*

## CONCLUSIONS AND CONTRIBUTIONS

*The evidence suggests that the explanatory relationship between variables proposed in this research is valid; however, it is more conclusive for the year 2020, given the behavior of the homicide variable.*

*The conditions that exacerbate the likelihood of committing homicide, that is, being net generator areas of offenders, are found in neighborhoods characterized by a high degree of marginalization for both periods.*

*This implies that, if the quality-of-life problems of these populations are not addressed, the areas established as generators and attractors will persist in the coming years.*

# REFERENCES


- Barajas Bustillos, H. A., y García De La Rosa, J. (2016). *Efectos de la eficiencia espacial de la dotación de bienes y servicios públicos en el bienestar de Ciudad Juárez en 2010*. Juárez, UACJ.
- Bass Zavala, S. (2013). “El crecimiento urbano en Ciudad Juárez, 1950-2000. Un acercamiento socio-histórico a la evolución desordenada de una ciudad de la frontera norte”. *ChihuahuaHoy2013*, 247–289. https://goo.su/In2yNUl
- Becker, G. S. (1968). Crime and punishment: An economic approach. Journal of political economy, 76(2), 169-217. https://doi.org/10.1086/259394
- Cervera Gómez, L. (2015). *Análisis Espacial de la Violencia en Ciudad Juárez*. [Tesis de doctorado]. Juárez, El Colegio de Chihuahua. https://goo.su/LP77nR
- Ehrlich, I. (1973). Participation in illegitimate activities: A theoretical and empirical investigation. Journal of political Economy, 81(3), 521-565. https://doi.org/10.1086/260058
- García De La Rosa, J., García Peña, A. K., y Buch Sánchez, E. (2018). *Perfil socioespacial de los homicidios en Ciudad Juárez para la intervención focalizada: Análisis desde la perspectiva de la víctima, el victimario y el lugar de los hechos*. Open Society Foundations. http://observatoriochihuahua.org/productos/evaluacion-e-investigaciones
- González, R. (1 de enero del 2024). “Cerró Ciudad Juárez el 2023 con mil 163 homicidios”. *El Heraldo de Chihuahua.* https://goo.su/8wJkxJq
- Loureiro, P. R. A., Cardoso De Mendonça, M. J., Silva Moreira, T. B., & Sachsida, A. (2009). “Crime, economic conditions, social interactions and family heritage”. *International Review of Law and Economics*, *29*(3), 202–209. https://doi.org/10.1016/j.irle.2009.01.002
- Payán, T. (2011). *Ciudad Juárez: la Tormenta Perfecta*. En N. Armijo Canto (Ed.), *Migración y seguridad: nuevo desafío en México* (pp. 127–144). Ciudad de México, Casede. https://goo.su/0Nlp6V